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Notes on al-Battānī's *Zīj*

from *Al-Battānī sive Albatenii Opus astronomicum* (Milan, 1899–1907)

translated from the Latin

Nallino's edition of al-Battānī's *Zīj* gives the Arabic text (Part III, 1899), a Latin translation of the canons with a commentary on them (Part I, 1903), and the tables in Latin with notes on them, a glossary and indexes (Part II, 1907). The canons themselves are translated into English from the Arabic in the English *Zīj* of this edition; this volume translates Nallino's own writing, which is in Latin and has not been translated before. It is made from the line-by-line transcription of his Latin in this edition (S03 and S04). Every sentence is translated, in Nallino's order; nothing is summarised or left out. His Arabic, Greek and Hebrew quotations stand in their scripts, as he prints them, with his translations of them in English. Proper names are given in a modern transliteration. In the headings of the commentary, «p.» and «l.» are the pages and lines of Nallino's Latin translation (Part I, pp. 1–150); «t. III» is the Arabic text, whose pages the English *Zīj* gives in its margin. Nallino's footnotes are numbered; the few notes of the translator are lettered. Nallino's own corrections in the «Addenda et emendanda» of Part II are followed and noted. The pages of Nallino's volumes are given in the margin.

The translation is in progress. This version holds the commentary on chapter 3, the commentary on chapter 4. It is AI-translated (Claude), 2026; no person has proofread it. Nallino's work is in the public domain; the translation is dedicated to the public domain (CC0 1.0).

Commentary

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On chapter 3

a) — pp. 9–10.

All the propositions on *chords* that the author gives are proved in the *Almagest* I, 9 (ed. Halma, vol. I, pp. 26–37). If we denote the radius by the letter R, they are expressed in our algebraic notation as follows:

(α) $\operatorname{chord}\frac{360^\circ}{6}$ or $\operatorname{chord} 60^\circ = R$

(β) $\operatorname{chord}\frac{360^\circ}{3}$ or $\operatorname{chord} 120^\circ = \sqrt{4R^2 - \operatorname{chord}^2\frac{360^\circ}{6}}$

(γ) $\operatorname{chord}(180^\circ - a) = \sqrt{4R^2 - \operatorname{chord}^2 a}$

(δ) $\operatorname{chord}\frac{360^\circ}{4}$ or $\operatorname{chord} 90^\circ = \sqrt{2R^2}$

(ε)[a] $\operatorname{chord}\frac{360^\circ}{10}$ or $\operatorname{chord} 36^\circ = \sqrt{R^2 + \left(\frac{2R}{4}\right)^2} - \frac{2R}{4}$

(ζ) $\operatorname{chord}\frac{360^\circ}{5}$ or $\operatorname{chord} 72^\circ = \sqrt{\operatorname{chord}^2\frac{360^\circ}{10} + R^2}$

(η) $\operatorname{chord}(a-b) = \frac{\operatorname{chord} a \operatorname{chord}(180^\circ - b) - \operatorname{chord} b \operatorname{chord}(180^\circ - a)}{2R}$

(ϑ) $\operatorname{chord}\frac{a}{2} = \sqrt{\frac{2R - \operatorname{chord}(180^\circ - a)}{2} \times 2R}$

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To find the chord of the sum of two arcs, Ptolemy (ed. Halma, vol. I, pp. 32–33), whom our author follows, has nothing like the rule (η), but goes by way of the chord of the supplement[1]; hence

(ι)[b] $\operatorname{chord}(180^\circ - [a+b]) = \frac{\operatorname{chord}(180^\circ - a)\operatorname{chord}(180^\circ - b) - \operatorname{chord} a \operatorname{chord} b}{2R}$

(κ) $\operatorname{chord}(a+b) = \sqrt{4R^2 - \operatorname{chord}^2(180^\circ - [a+b])}$

Almost all the historians of mathematics describe the way by which Ptolemy came to compute the chord of 1°; of the most recent, one may consult M. Cantor, *Vorlesungen über Geschichte der Mathematik*, Leipzig 1880 ff., vol. I, pp. 353–355 (or 390–392 of the second edition, of 1894); R. Wolf, *Handbuch der Astronomie, ihrer Geschichte und Litteratur*, Zürich 1890–93, vol. I, pp. 164–165, § 61*b*; A. von Braunmühl, *Vorlesungen über Geschichte der Trigonometrie*, Leipzig 1900 ff., vol. I, p. 21. The error, which F. Woepcke determined with Lagrange's interpolation formula (*Discussion de deux méthodes arabes pour déterminer une valeur rapprochée de sin 1°*, in the Journal de Mathématiques pures et appliquées published by J. Liouville, vol. XIX, 1854, p. 165), is 0.00000009, that is, less than 1/10,000,000; it therefore becomes noticeable only in the thirds,

[1] Delambre, *Astron. ancienne*, vol. II, p. 39, is mistaken (as Schiaparelli points out to me) in ascribing to Ptolemy the rule

$$\operatorname{crd}(p+q) = \frac{\operatorname{crd} p \operatorname{crd}(180^\circ - q) + \operatorname{crd} q \operatorname{crd}(180^\circ - p)}{2R}.$$

[*Translator:* The print has – between the two products; Nallino's Addenda et emendanda (Part II, p. XXI) correct it to +, which gives the chord of the sum.]

which Ptolemy leaves out of his table of chords. — A value of the sine of 1° nearer the truth was found by Abū l-Wafāʾ (940–998), whose error shows only in the fifths, that is, in the ninth decimal place (sin 30′ = $0^p\,31'\,24''\,55'''\,54^{IV}\,55^{V}$ instead of $54^{V}\,47^{VI}\,16^{VII}$); see Woepcke, *Recherches sur l'histoire des sciences mathématiques chez les Orientaux*, in the Journal Asiatique, 5th series, vol. XV, 1860, pp. 286–303; Cantor, I, 641–642 (= 703 of the second edition); von Braunmühl, I, 56–57. — According to what Mīram Čelebī (d. 931 H = 1524–25) has handed down to us in his commentary on Ulugh Beg (in Sédillot's French translation, pp. 69–77, note), other, early Arabs, whose names are not known, carried Ptolemy's method through and found a value of the sine of 1° that exceeds the truth by less than 1/100,000,000; for the error is 37/416 (or about 1/11) of Ptolemy's error, since they make sin 1° = $1^p\,2'\,49''\,43'''\,13^{IV}\,5^{V}\,30^{VI}$ instead of $11^{IV}\,14^{V}\,44^{VI}$. But had they not deliberately neglected certain small fractions in their computations, they could have found a far better number, whose error would have been 1/64 of Ptolemy's. See Woepcke, *Discussion de deux méthodes* etc., pp. 155–157, 159–167. — Lastly, Ghiyāth al-Dīn[1] Jamshīd al-Kāshī, in whom other Arabs had probably gone before him, computed sin 1° = $1^p\,2'\,49''\,43'''\,11^{IV}$ correctly and most elegantly, by solving the equation of the trisection of the angle approximately, that is, by expanding the equation of the third degree in a series; his method, handed down to us by Mīram Čelebī, loc. cit., pp. 77–83, was explained by Woepcke, *Discussion*, pp. 167–176, 301–303. See also H. Hankel, *Zur Geschichte der Mathematik im Alterthum und Mittelalter*, Leipzig 1874, pp. 290–293; Cantor, I, 671–672 (= 737 of the second edition); von Braunmühl, I, 73–75.

b) — pp. 10–11.

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The technical word *jayb*, by which the Arabs denote the sine, al-Battānī never mentions, although it was already in common use at the beginning of the ninth century, as appears from the unpublished Latin translation of the astronomical work of Muḥammad ibn Mūsā al-Khwārizmī, made by Adelard of Bath in the twelfth century. For there the Arabic name *elgeib* is kept (with the Arabic article *al* or *el* put before it), and «*elgeib elmustewi* or plane» [*al-jayb al-mustawī*], that is, the right sine, and «*elgeib elmacus* or diminished»[2], that is, the versed sine, are named. Likewise Ḥabash, a contemporary of al-Battānī, always uses the word *jayb*. But our author would not take up the word, which is of foreign origin and wholly alien to Greek usage. Not a few historians of mathematics, then, and among the best, are wrong when they ascribe the use of the word *jayb* to our author and think that Plato of Tivoli brought the word *sinus* to Europe with his translations; of the more recent I shall cite only Hankel, p. 280; Wolf, *Handb. der Astron.*, I, 166, § 62*a*; Cantor, I, 632 (= 693 of the second edition), II, 101 and 249. The old Kaestner, however, had already corrected the error, *Geschichte der Mathematik*, Göttingen 1796, vol. I, p. 522; see now M. Koppe, *Die Behandlung der Logarithmen und der Sinus*, Programm des Andreas-Realgymnasium, Berlin 1893, p. 32; von Braunmühl, I, 50. — The word *sinus* occurs only once in Plato of Tivoli's translation: «si autem per has cordas [*sic*] versas arcus scire desideras, si chorda quam habueris minus 60 fuerit, eam de 60 minue, et residui arcum scito, quem de 90 minues, et quod remanserit, erit arcus *sinus* versi» («but if you wish to know the arcs from these versed chords: if the chord you have is less than 60, take it from 60 and find the arc of the remainder; take that from 90, and what remains will be the arc of the versed *sinus*»; fol. 7v of the Nuremberg edition; the passage corresponds to p. 11, ll. 28–30 of my translation). But, as von Braunmühl, I, 50, note 3, rightly judges, the word must be taken for a gloss of Regiomontanus that has crept into the text by mistake.

[1] Some indeed call him, wrongly, ʿAtāb al-Dīn, reading عتاب ʿAtāb for غياث Ghiyāth. He died about the year 840 H, 1436–37, as H. Suter, *Die Mathematiker und Astronomen der Araber*, Leipzig 1900, pp. 173–4, no. 429, rightly established; to his arguments one may add that the little work on finding the sine of 1° (which still survives in the Cairo Library) is already praised by Qāḍīzāde al-Rūmī, who died between 840 and 850 H, as Mīram Čelebī relates.

[2] Adelard's error; *maʿkūs* = versed, inverted. The passage was cited by Chasles, *Recherches sur l'astronomie ancienne*, in the Comptes rendus de l'Académie des Sciences, vol. XXIII, session of 2 November 1846, p. 850.

Since chords are hardly ever used in trigonometric computations, it is no wonder that al-Battānī called the halves of the chords of the double arcs by the name of chords; a similar usage is found among the Indians as early as the fourth century, of whom one reads the following in the notes of Burgess and Whitney on the book entitled *Sūrya-siddhānta*: «In this passage, the sine is called *jyārdha*, 'half-chord'; hereafter, however, that term does not once occur, but *jyā* 'chord' (literally 'bowstring') is itself employed, as are also its synonyms *jīvā*, *māurvikā*, to denote the sine. The usage of Albategnius is the same» (*Translation of the Sūrya-Siddhānta, a text-book of Hindu astronomy, by* E. Burgess, note on ch. II, verse 28, in the Journal of the American Oriental Society, vol. VI, New Haven 1860, p. 201). — Nor should it be passed over that among the Arabs the name of the whole *chord* too comes from the usage of the Indians, as the passage just quoted shows; among the Greeks the chord was called ὑποτείνουσα, «subtending», ἡ ἐν τῷ κύκλῳ εὐθεῖα, «the straight line in the circle», εὐθεῖα ἐν κύκλῳ μὴ διὰ τοῦ κέντρου οὖσα, «a straight line in the circle not passing through the centre», εὐθεῖα ἐλάττων τοῦ διαμέτρου, «a straight line smaller than the diameter».

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The origin of the technical word *jayb* was explained satisfactorily by Munk, whom F. Woepcke, *Essai sur la propagation des chiffres indiens*, in the Journal Asiatique, 6th series, vol. I, 1863, p. 478, note, and almost all later writers have followed. From the Indian word *jīvā*, chord and sine, which I mentioned a little above, the Arabs, who have no letter *v*, made *jīb* جيب; for the Sanskrit *j* is pronounced like the Arabic *j*. But since *jīb* has no meaning in Arabic, while the letters جيب can also be read *jayb* without any change to the writing (for in ordinary writing the small signs that show the true pronunciation of the vowels are almost always left out), the Arabs preferred this form of the name, which in their language means not only «the round opening of a garment through which a man's neck and head pass», but also «the fold or pocket made in a garment for carrying small things»; occasionally, but rarely, it also means a man's bosom itself. We have already seen that Adelard of Bath kept the Arabic name *elgeib* in his translations; but his contemporary Gerard of Cremona (1114–1187), in translating the astronomy written by Jābir ibn Aflaḥ, used the word *sinus*, which came into our languages by this route, as Koppe, loc. cit., rightly observed[1]. J. Ruska, *Zur Geschichte des «Sinus»* (in the Zeitschrift für Mathematik und Physik, vol. XL, Leipzig 1895, historisch-literarische Abtheilung, p. 128), cites two Syriac writers of the thirteenth century[2] in whom the technical word *jayb* is always rendered by the Syriac *ʿūbā* ܥܘܒܐ, which means a man's bosom; but he is wrong when he writes: «Die Uebersetzung durch *sinus* ist demnach als richtig durch die syrische Belege gesichert. Die verständigere Uebersetzung aber, welche einen Mann voraussetzt, dem der mathematische Inhalt und dessen klarer Ausdruck wichtiger ist als pedantische Wörtlichkeit, lautet natürlich *chorda* . . . . . Beide Uebersetzungen finden sich vor: die zweite, wie Koppe nachgewiesen hat, bei Plato von Tivoli» («The translation by *sinus* is thus confirmed as correct by the Syriac evidence. The more sensible translation, however, which presupposes a man to whom the mathematical content and its clear expression matter more than pedantic literalness, is of course *chorda* . . . . . Both translations occur: the second, as Koppe has shown, in Plato of Tivoli»). For Plato of Tivoli writes *chorda* not from any striving after a more rational translation (how foreign that was to him and to the other translators of his age, their translations show plainly enough everywhere), but because al-Battānī uses this very word.

That the Arab mathematicians learned from the Indians the terms unknown to the Greeks — the chord, the sine, the right sine (Sanskrit *kramajyā*), the versed sine (Sanskrit *utkramajyā*), the sagitta (Sanskrit *śara* or *ishu*) — can hardly be doubted. To this one may add what the Arab writers relate of the Indian book entitled *al-Sindhind* (that is, the Sanskrit *Brahma-sphuṭa-siddhānta*), which was brought to the caliph at Baghdad in the year 156 H, 773. When all this is weighed, the

[1] But from the fact that it was Gerard of Cremona, not Plato of Tivoli, who spread the use of this word among us, I do not see how Koppe can call into doubt the origin of the Arabic *jayb* that Munk explained.

[2] They belong, then, to a late age not only of Syriac but also of Arabic letters, and not, as von Braunmühl, I, 50, note 1, writes, «in den ersten Zeiten des Mittelalters», to the first times of the Middle Ages.

history of the trigonometric functions can be told as follows. The Indians, moved above all by their use of orthogonal projections (von Braunmühl, I, 31–32), used right and versed sines instead of the chords of the Greeks; they computed them for the twenty-fourth parts of the quadrant, that is, for every 3° 45′, taking a radius of 3438 minutes. Once they had received the first notion of sines from the Indians, and had come to know Ptolemy's books at about the same time, the Arabs rightly understood that it was more convenient and more exact to construct new sines by halving Ptolemy's chords (which are computed for every half degree and a radius of sixty parts). This had already been done before al-Battānī.

c) — p. 11.

The *interpolation* for finding the sines of arcs that fall between two consecutive arcs of the table is the same as the one found in Ptolemy for computing chords: the change of the sines and of the chords is taken to be proportional to the change of the arcs within the interval of the table, half a degree. What Cantor, I, 355 (= 392 of the second edition), writes of Ptolemy, «ein Beweis für diese angenommene Proportionalität in engem Bereiche ist dagegen nicht vorhanden» («a proof of this proportionality, assumed within a narrow range, is however not given»), may rightly be said also of al-Battānī, of Ḥabash (fol. 18v) and of the other Arabs. More exact interpolations they used only in constructing tables. — Tables of sines for every minute of the quadrant, computed by Ibn Yūnus, survive in the Arabic codex 5752 of the Royal Library of Berlin; Ulugh Beg also computed others.

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Versed sines, whose use the Indians knew before the Arabs, the author defines:

$$\text{versed sine } a = R - \cos a.$$

He thought it superfluous to construct a table of them, which on the other hand we find in Ḥabash. At the foot of the page two formulas are given; one is

(α) $\quad \cos a = \sqrt{R^2 - \sin^2 a},$

which follows from the theorem of Pythagoras and, with $R = 1$, corresponds to our familiar rule

$$\cos a = \sqrt{1 - \sin^2 a}.$$

The other,

(β) $\quad \sin \dfrac{90^\circ - a}{2} = \sqrt{30^p\,(R - \sin a)},$

comes from the rule of al-Battānī and of Ptolemy mentioned in note *a* (p. 153),

$$\text{chord } \frac{a}{2} = \sqrt{\frac{2R - \text{chord}\,(180^\circ - a)}{2} \times 2R},$$

which, when sines are put for chords and $R = 60^p$ is taken, becomes

$$\sin \frac{a}{2} = \sqrt{30^p\,(R - \cos a)}.$$

Why al-Battānī took the complement of *a* in place of *a*, I do not know.

On chapter 4

a) — p. 12.

Of the Arabs who measured the *obliquity of the ecliptic* before the year 1000, Ibn Yūnus relates the following in the eleventh chapter of his work[1]: «Between Ptolemy and the authors of the *Tested Tables* (*al-mumtaḥan*) I know of no observations of the obliquity except one made a little after[2] the year 160 of the Hijra [which began on 18 October 776], whose author gives the obliquity as 23° 31′. As for the value found by the authors of the *Tested Tables*[3] in observations close to our own age: Yaḥyā ibn Abī Manṣūr, Sanad ibn ʿAlī, al-ʿAbbās ibn Saʿīd al-Jawharī and a group of learned and eminent men (وطايفة من اهل العلم والفضل) investigated it in [the quarter called] al-Shammāsiyya of the city of Baghdad, in the time of [the caliph] al-Maʾmūn, and found 23° 33′, as Muḥammad ibn Mūsā al-Khwārizmī in his astronomical work (في زيجه) and Muḥammad ibn Kathīr al-Farghānī in his book on the making of the astrolabe report on their authority; and the matter is well known[4]. Those who observed at Damascus after the death of Yaḥyā ibn Abī Manṣūr, with the instrument that al-Maʾmūn had ordered to be made when he set out for the land of the Byzantines (بالالة التي امر باتخاذها المامون حين توجـه الى بـلاد الروم) — they are Khālid ibn ʿAbd al-Malik al-Marwarrūdhī, Abū l-Ṭayyib Sanad ibn ʿAlī, ʿAlī ibn ʿĪsā[a] al-Asṭurlābī and others — report that they found the greatest declination of the ecliptic to be 23° 33′ 52″; and the measurement made with this instrument was in the year 201 of Yazdajird» [which began on 27 April 832][5]. I 158

«Abū l-Ḥasan Thābit ibn Qurra says: I found ancient operations from before Ptolemy which show (تدل) that the greatest declination is 23° 35′[6]. Muḥammad ibn Jābir ibn Sinān al-Battānī relates that he found 23° 35′ by his observations. The noble and eminent Abū l-Qāsim ʿAlī ibn al-Ḥusayn ibn Muḥammad ibn ʿĪsā, known as Ibn al-Aʿlam, a man of learning and of outstanding intelligence, says that he measured the greatest declination and, taking the utmost care in the measurement, found 23° 34′ 2″. Abū l-Ḥasan al-Ṣūfī ʿAbd al-Raḥmān ibn ʿUmar relates that he measured the greatest declination and, having carried out the measurement with the greatest accuracy, found 23° 33′ 45″[7]; that man held an eminent place in the science of geometrical proofs. Help comes from God.»

Ibn Yūnus, ch. IV (in Caussin, p. 120), says that the value 23° 35′ was chosen in the *Tested Tables*; and al-Farghānī reports the same in ch. V (p. 20 of Golius's edition, fol. 121r of the Paris codex 2504). But Ibn Rusta, who copies the whole of this chapter of his contemporary al-Farghānī, has

[1] P. 222 of the Leiden codex (of which the Paris codex no. 2495 is a most careful copy, keeping the same division of the pages). I compared the passage with the Paris codex 2496 (which gives selected chapters of Ibn Yūnus's work and is written more carelessly), fol. 25r.

[2] What I have rendered «a little after» is in Arabic نيف, that is, «more than two but fewer than ten» [years after].

[3] This is the astronomical work that the astronomers of the caliph al-Maʾmūn (reigned 813–833) compiled from their observations under the direction of Yaḥyā ibn Abī Manṣūr.

[4] Ibn Yūnus himself confirms this in the introduction to his work (in Caussin, p. 56), saying that in the year 214 H (which began on 10 March 829), 198 of Yazdajird (which began on 28 April 829), Yaḥyā ibn Abī Manṣūr, al-ʿAbbās ibn Saʿīd al-Jawharī, Sanad ibn ʿAlī and others found 23° 33′ in the city of Baghdad. — Al-Zarqālī, in the unpublished Latin translation of the *Canons* or *Toledan Tables*, and the Jew Abraham ibn Ezra (in the second half of the twelfth century) say that Yaḥyā ibn Abī Manṣūr found 23° 33′ 30″; see Delambre, p. 177 (whose codex of the Toledan Tables gave the name in a corrupt form, «Almeon fils d'Albumazar»), and M. Steinschneider, *Études sur Zarkali*, in the Bullettino di bibliografia e di storia delle scienze matematiche e fisiche, vol. XX, 1887–88, p. 87 (where a codex of the Tables is cited that has the name «Jahiben filius Albumasaris»).

[5] He relates the same in the introduction (Caussin, 56); the year there is 217 H (which began on 7 February 832), 201 of Yazdajird.

[6] But in his own works Thābit chose 23° 33′, as appears from Ḥabash, fol. 90v, and Delambre, p. 73.

[7] The Paris codex 2496 wrongly writes 23. 35. 45. 38.

[a] The print has ʿAlī ibn Mūsā; Nallino's Addenda et emendanda (Part II, p. XXI) correct it to ʿAlī ibn ʿĪsā.

23° 33′ for 23° 35′[1]; and 23° 33′ is also read in the Latin translation of al-Farghānī made by John of Seville and printed at Nuremberg in 1537 (fol. 5r). This seems the more probable when one compares what Ibn Yūnus himself relates of the measurement made by the authors of the *Tested Tables*, and what Rabiçag of Toledo writes in his *Libro de las armellas*[2], which is drawn wholly from Arabic sources: «Et sabet que en este nuestro tiempo segund el nostro rectificar. bien se paresció por los rectificamentos antigos et los nuevos. que lo que ha entre ellos [that is, the interval between the pole of the ecliptic and the pole of the equator] non es uno todauia. Et esto es porque Abrachis [= Hipparchus] falló la declinacion que era .XXIII. grados et .LI. menudos. et falláronla los rectificadores [that is, the authors of the Tested Tables] en el tiempo de Memun Abdalla. fiio de Haron Arraxit .XXIII. grados et XXXIII. menudos. Et fallóla Mohamat. fiio de Aben el Bateni [= Muḥammad ibn Sinān al-Battānī] .XXIII. grados et .XXXV. menudos» («And know that in this our time, by our rectifying, it has appeared clearly from the ancient and the new rectifications that what lies between them is not always one. And this is because Abrachis found the declination to be 23 degrees and 51 minutes; and the rectifiers found it in the time of Memun Abdalla, son of Haron Arraxit, to be 23 degrees and 33 minutes; and Mohamat, son of Aben el Bateni, found it to be 23 degrees and 35 minutes»). — Thābit ibn Qurra says the same in the book *On the motion of the eighth sphere, or of access and recess*, which survives in two different Latin translations in the Vatican Library; one (cod. Lat. 4275, f. 84v, col. I) says: «Et eius [that is, of the whole obliquity] summa est 24 graduum secundum quod translatum est ab Indis, et similiter reperit Ptholomeus 23 g. et 51 minut., et consideratores [that is, اصحاب الممتحن, the authors of the Tested Tables] invenerunt in tempore Maymonis 23 g. et 33 m.» («and its amount is 24 degrees according to what has been handed down from the Indians, and likewise Ptolemy found 23 degrees and 51 minutes, and the observers found in the time of al-Maʾmūn 23 degrees and 33 minutes»); the other (cod. Lat. 4083, f. 56v, col. II): «pervenit talis declinatio ad 24 gradus secundum quod ab Indis traditum fuit. Ptolomeus quoque invenit eam 23 graduum et 51 mt. Alii autem inspectores post Ptolomeum in partem Mamonis eam 23 graduum 33 minutorum invenerunt»[a] («this declination reached 24 degrees according to what was handed down by the Indians. Ptolemy too found it to be 23 degrees and 51 minutes. But other observers after Ptolemy, on the side of al-Maʾmūn, found it to be 23 degrees 33 minutes»). — Lastly, it should not be passed over that Thābit ibn Qurra, as we saw in note 1, chose 23° 33′; and so did Qusṭā ibn Lūqā, who died about 910 or 912, in his book on the construction and use of the celestial globe[3]. — They would hardly have chosen 23° 33′ had they not found this value in the *Tested Tables*.

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Of those who flourished after the authors of the *Tested Tables*, down to the last years of al-Battānī, we know that the family of the Banū Mūsā[4], al-Nayrīzī[5] and the family of the Banū Amājūr[6] found the obliquity to be 23° 35′ by observation. So did, among later astronomers, Abū l-Wafāʾ, in the city of Baghdad, in the year 987 (on the evidence of Delambre, p. 156), and Ibn Yūnus in the year 391 H (which began on 30 November 1000; ch. XI, in Caussin, p. 238; more exactly he had found 23° 34′ 52″). — Many others adopted this obliquity of 23° 35′ in their tables, such as Ḥabash (a

[1]Ibn Rusta, *Kitâb al-Aʿlâk an-Nafisa VII*, edited by M. J. de Goeje, Leiden 1889, p. 16 [= Bibliotheca Geographorum Arabicorum, vol. VII].

[2]Book I, ch. IV = *Libros del saber de astronomia del Rey D. Alfonso X de Castilla, copilados, anotados y comentados por D. Manuel Rico y Sinobas*, Madrid 1863–67, vol. II, p. 6. It belongs to the second half of the thirteenth century.

[3]Ch. XXXIV of the Spanish translation (*Libros del saber de astronomia*, vol. I, p. 187), ch. XXX of the unpublished Hebrew translation (in Baldi, *Vite di matematici arabi con note di* M. Steinschneider, Rome 1874, p. 70, note).

[4]Ibn Yūnus, ch. V, in Caussin, p. 152. — One of these Banū Mūsā died in the month of Rabīʿ I 259 H = January 873; on them see Suter, *Die Mathematiker und Astronomen der Araber und ihre Werke*, Leipzig 1900, pp. 20–21, no. 43 (= Abhandlungen zur Geschichte der mathematischen Wissenschaften, part X; supplement to the 45th year of the Zeitschrift für Mathematik und Physik).

[5]Ibn Yūnus, ch. IV, in Caussin, p. 120. — He died about the year 310 H (922–23); see Suter, p. 45, no. 88.

[6]Ibn Yūnus, ch. V, in Caussin, p. 152. — They observed in the years 272–321 H (885–933); see Suter, pp. 49–50, no. 99.

[a]The print has «quos» for «quoque» and «tempore» for «partem»; Nallino's Addenda et emendanda (Part I, p. LXXII) correct both.

contemporary of al-Battānī), al-Kharaqī in his *Muntahā al-idrāk*[1], the Hebrew-Spanish Abraham bar Ḥiyya, writing in 1135[2], and his contemporary ʿAbd al-Raḥmān al-Khāzinī[3]; likewise Naṣīr al-Dīn al-Ṭūsī in the books he wrote before the *Īlkhānī Tables*[4], the Greek Chrysococces in his *Persian Tables*[5], and lastly, at the beginning of the fourteenth century, al-Jaghmīnī (as appears from pp. 264 and 265). Barhebraeus, or Abū l-Faraj, in his Syriac Astronomy written in 1279, uses round numbers, and so in places he gives the obliquity as 24°[6]; but where he wants more exact numbers he takes it as $23^\circ\ 35'$, as can easily be inferred from his computations of the right and the oblique ascensions (pp. 165–167 of the Syriac text).

Abū ʿAlī al-Ḥasan (that is, Aboul Hhassan), ch. XXIII, pp. 175 and 176, says that the obliquity varies between the limits $23^\circ\ 33'$ and $23^\circ\ 53'$; but in ch. XXIV, pp. 180–187, and in his tables, he adopted $23^\circ\ 35'$.

It should be noted that an error of two minutes in fixing the obliquity is of little consequence in the computations derived from it; for in the right ascensions it can produce a discrepancy only in the seconds of arc[7]; in the oblique ascensions the greatest difference never reaches three minutes of arc (especially since the latitudes that the Arabs had to use were always smaller than 45°); and lastly, in computing the greatest half arc of the day of the sun in the lands under Arab rule, the difference can never exceed ten seconds of an hour. Such differences could be neglected in those times.

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So that one may judge with what care the Arabs observed the greatest declination of the ecliptic, I have made a small table of the chief observations, using in my computations Bessel's formula, which for t years before 1750 is

$$\varepsilon = 23^\circ\ 28'\ 18.0'' + 0.48368''\,t - 0.00000272295''\,t^2.$$

With an asterisk I mark the years of the observations that are certain; where the year is not known, I have put the epoch of the author's tables, or the year that can be assumed approximately. I have given the errors to the second; but everyone knows that these numbers cannot always be taken absolutely, since sometimes the year of the observation is not known for certain, or the Arab observer deliberately neglected the seconds[8].

[1] Laurentian codex (Orient.-Palat. 110), f. 40r (maqāla I, bāb XIV, faṣl I).

[2] *Sphaera mundi auctore Rabbi Abrahamo Hispano filio R. Haijae. Osw. Schreckenfuchsius vertit in linguam Latinam, Seb. Munsterus illustravit annotationibus.* Basel 1546, ch. X.

[3] Vatican Arabic codex 761, f. 10v and in many places.

[4] As in the *Tadhkira*, Laurentian codex (Palat.-Orient. 271), f. 38v (bāb II, faṣl IV).

[5] In Ismael Bullialdus, *Astronomia Philolaica*, Paris 1545, part II, p. 224. — Bullialdus thinks that he flourished in the first half of the thirteenth century (loc. cit., p. 213); but Krumbacher, *Geschichte der Byzantinischen Litteratur*, 2nd ed., Munich 1897, says that he lived about the year 1346.

[6] This value was known to the Greeks before Eratosthenes; the Indians always used it. — See also below, p. 161, note 3.

[7] [The greatest difference is in the octants, that is, at the right ascensions 45°, 135°, 225°, 315°, where an error of one minute in fixing the obliquity produces an error of $0.217'$ (or very nearly $13''$) in the right ascension. — Schiaparelli]

[8] The small tables that, after Lalande (I think), circulate among our writers (the most careful Houzeau and Wolf not excepted) contain errors whose origin I do not know; for they ascribe to Eratosthenes $23^\circ\ 45'\ 7''$, to al-Battānī $23^\circ\ 35'\ 41''$, to Ibn Yūnus $23^\circ\ 34'\ 26''$, to Ulugh Beg $23^\circ\ 31'\ 48''$, to Abū ʿAlī al-Ḥasan (Aboul Hhassan) $23^\circ\ 33'\ 45''$!

Observers	*Place of the observation*	*Year*	*Obliquity observed*	*True (mean) obliquity*
Eratosthenes	Alexandria	230 BC	23° 51′ 20″	23° 43′ 45.01″
Hipparchus[1]	Rhodes	130 BC	23 51 20	23 42 56.62
The authors of the Tested Tables[2]	Baghdad	* 829	23 33	23 35 41.16
Other astronomers of the caliph al-Maʾmūn	Damascus	* 832	23 33 52	23 35 39.71
Banū Mūsā	Baghdad	860	23 35	23 35 26.32
al-Battānī	al-Raqqa	880	23 35	23 35 16.74
Ibn al-Aʿlam	Baghdad?	960	23 34 2	23 34 37.92
ʿAbd al-Raḥmān al-Ṣūfī	Baghdad?	965	23 33 45	23 34 35.50
Abū l-Wafāʾ	Baghdad	* 987	23 35	23 34 25.46
Wayjan ibn Rustam al-Kūhī[3]	Baghdad	* 988	23 51 1	23 34 24.98
Ibn Yūnus	Cairo	* 1001	23 34 52	23 34 18.74
Alfonsine Tables[4]	Toledo	1250	23 32 29	23 32 19.16
Naṣīr al-Dīn al-Ṭūsī[5]	Marāgha	1270	23 30	23 32 9.49
Ibn al-Shāṭir	Damascus	1363	23 31	23 31 24.78
Ulugh Beg	Samarqand	1437	23 30 17	23 30 49.13

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The author's words, p. 12, ll. 14–24, fully confirm what I had already said in note 9, p. 5: that al-Battānī could by no means have inferred a *continuous* decrease of the obliquity of the ecliptic from the uncertain observations of the Greeks. How could he have argued it, when from Eratosthenes to Theon, that is, over an interval of about 600 years, all the Greeks always say they found 23° 51′ 20″?

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To the later Arabs, who could compare many good observations with one another, the hypothesis of a continuous decrease was not unknown, although the small discrepancies could have been put down to errors of the observers and of the instruments. Indeed, we read the following in the astronomical book entitled the Sanjarī (*al-zīj al-sanjarī*), which Abū Manṣūr ʿAbd al-Raḥmān al-Khāzinī wrote about the year 520 H (1126)[6]: «In [fixing] the obliquity of the ecliptic there are many discrepancies, about whose cause opinions differ. The most likely of all is that they arise only from the refraction of the rays (وغالب الظنّ ان سببه انعطاف الشعاع دون سايره), so that, when the latitude of two observers is one and the same, the same obliquity appears to both. The greatest declination is 24° 0′ with the Indians, 23° 51′ with Ptolemy and Hipparchus, 23° 35′ with the authors of the Tested [Tables] and al-Battānī, 23° 33′ with some of the moderns (بعض المحدثين). Because of these discrepancies, and because of the irregularity in the motion of the fixed stars, some thought fit (ظهر لبعضهم) to posit a motion in the space between the two spheres, that of the universe and that of the fixed stars, which would complete the whole circle every 1285 (١٢٨٥) solar years, and whose poles, like the poles of the fixed stars too, would not lie on the axis of the universe; so that the poles of the signs of the zodiac would approach the equators or recede from them, and the motion of the fixed stars would be quickened or slowed, in a way that it would take too long to set out now. But since the obliquity is 23° 35′ with most [authors], we shall use this value in our reckonings, as one rightly accepted (استعملناه في الاعتبار مُسلا)». — Naṣīr

[1] Ptolemy, in the year 140, keeps the same value; his error is therefore + 10′ 10″. The error of Theon, who wrote about the year 360, is + 12′ 1″.

[2] If 23° 33′ 30″ is admitted, as some say (see p. 157, note 4), the error becomes − 2′ 11″. [*Translator:* The print refers to «p. 6, note 3»; Nallino's Addenda et emendanda (Part II, p. XXI) correct the reference to p. 157, note 4.]

[3] According to the passage of Ibn al-Qifṭī that Casiri, *Bibliotheca Arabico-hispana Escurialensis*, Madrid 1760–70, vol. I, p. 441, and Sédillot, in the preface to the text of Oloug Beg, pp. LII–LIII, cite. The codex of Ibn al-Qifṭī kept at Rome in the Library of the Accademia dei Lincei gives the same reading.

[4] So in many places in the *Libro de las taulas alfonsies* (*Libros del saber de astronomia*, vol. IV) and in the *Libro del quadrante para rectificar* (ibid., vol. III, pp. 296–297).

[5] According to Sédillot's note on the text of Oloug Beg, p. 63; the passage he cites is found on f. 105r of the Florence codex of these same *Īlkhānī Tables*. The first declinations (ff. 120r–121r) and the second (ff. 121v–122v) are computed with this value.

[6] Vatican Arabic codex no. 761, in the third chapter (*bāb*) of the preface, f. 10v.

al-Dīn al-Ṭūsī, in his Memorandum of what is chiefly to be remembered in astronomy[1], relates the following: «The total declination found by the ancient and by the more recent observations is not one and the same; the value the ancients found is greater than the one the more recent have measured. What a more recent observer found is sometimes taken to be less than what an earlier one found, the greatest value found not reaching 24°[2], while the least does not go below 23° 1/2 + 1/(2 × 10) [= 23° 33′][3]. Most (الجمهور) set it at 23° 1/4 + 1/3 [= 23° 35′]. Because of these discrepancies some have thought that the ecliptic moves in latitude and approaches the equator; if this were true, another sphere would have to be posited that carries the zodiac with such a motion. If the ecliptic moved, it could complete the circle; or it could not complete it, but reach a certain limit from which it would afterwards turn back. Any such limit could lie after the two falls of the ecliptic onto the equator, or at the time of the second fall, or between the two falls; and this last [in three ways, namely] after half the circle has been completed, or exactly at the half circle, or before the semicircle is completed. Lastly, the ecliptic could also not reach the space between the two falls, if it is thought to turn back either at the time of its first fall onto the equator or before it. There are, then, eight hypotheses. By the first five, the two halves of the zodiac, the northern and the southern, would exchange places, together with their properties; and by three of these [five], each half of the zodiac would [at times] fall onto each half of the equator. By the first seven hypotheses, a half of the ecliptic would coincide with the half of the equator nearest to it; whenever this happened, days and nights would be equal in all places of the earth, and the different seasons of the year would fail. By the last [that is, the eighth] hypothesis[4], however, there would follow only an increase or a decrease in the altitudes [of the sun] and in the lengths of the days and the nights».

Quṭb al-Dīn Maḥmūd al-Shīrāzī, who died in the month of Ramaḍān 710 (February 1311), says in his *al-Tuḥfa al-shāhiyya* (ff. 5v–6r) that no definite rate of decrease is found among the different values of the obliquity measured at different times.

Those who have written the history of astronomy say that in the second half of the sixteenth century some first recognized the continuous decrease of the obliquity, such as Ignazio Danti in his *Trattato dell'Astrolabio*, Florence 1569, though many others of that age called it into doubt; that Tycho Brahe (1546–1601) inferred it from the changed latitudes of the stars; and that, lastly, Euler (1707–1783) proved it by the laws of celestial mechanics.

b) — p. 12, ll. 27–32.

The rule for finding the **declination** of the *sun*, or of any degree of the ecliptic, follows from Ptolemy's construction, *Almag.* I, 12 (ed. Halma, vol. I, pp. 56–58). Let ε be the obliquity of the

[1] *Al-Tadhkira fī ʿilm al-hay'a*, in the fourth chapter (*faṣl*) of the second section (*bāb*), ff. 38r–39r of the codex of the Laurentian Library, which also contains the commentary of Qāḍīzāde. — A French summary of the passage is given by Carra de Vaux, *Les sphères célestes selon Nasîr-eddîn Attûsî*, at the end of P. Tannery's *Recherches sur l'histoire de l'astronomie ancienne*, Paris 1893, p. 346.

[2] The Indian astronomers always have 24°. — Qāḍīzāde remarks on this passage: «Some think that the sages of the Indians found it [24°], and that this same value was very well known in the time of Euclid, which is why [Euclid] in his book, after computing the side of the hexagon inscribed in a circle, derived the side of the pentadecagon inscribed in a circle, since 24 is the fifteenth part of the circumference». — So too Quṭb al-Dīn Maḥmūd al-Shīrāzī, *al-Tuḥfa al-shāhiyya*, f. 5v: «Some think that all the Indians agree on this value [24°], and that 24° was the obliquity in the time of Euclid; which is why he derived the side of the pentadecagon in his book, since 24 is the fifteenth part of the circumference». — Strange enough that some saw the obliquity of the ecliptic foreshadowed in the fifteenth and sixteenth propositions of the fourth book of Euclid, on inscribing regular hexagons and pentadecagons in a circle! [*Translator:* Nallino's Addenda et emendanda (Part II, pp. XXI–XXII) trace this story to Heron of Alexandria's commentary on Euclid, as quoted by al-Nayrīzī, and cite Proclus and, after Eudemus, Theon of Smyrna, who give the distance between the two poles as the side of the fifteen-sided polygon, 24°; the Addenda are translated with Part II.]

[3] Qāḍīzāde: «So at the time this book was written; but afterwards the author himself, with the new observations that he set up [in the city of] Marāgha, found 23° 30′, which is the least value found so far». And in fact Naṣīr al-Dīn adopted the obliquity 23° 30′ in his *Īlkhānī* tables.

[4] By which the ecliptic is thought never to coincide with the equator.

ecliptic, δ the declination of the given degree, L the longitude of the sun or of the given degree; by the well-known theorem of Menelaus Ptolemy proves

$$\frac{\text{chord } 180^\circ}{\text{chord } 2\varepsilon} = \frac{\text{chord } 180^\circ}{\text{chord } 2\delta} \times \frac{\text{chord } 2L}{\text{chord } 180^\circ}.$$

Since chord 180° is the diameter (2R), if sines are used instead of chords, there results

$$\frac{2R}{2\sin\varepsilon} = \frac{2R}{2\sin\delta} \times \frac{2\sin L}{2R},$$

whence al-Battānī's formula[1]

$$\sin\delta = \frac{\sin L \ \sin\varepsilon}{R}.$$

c) — p. 13, ll. 15–18.

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Nothing like this is read in Ptolemy, but it is found in Theon's *Canon*, ed. Halma, vol. I, p. 54: «For when the sun is between 1° and 90° we shall say that it is descending towards the north [*sic*, βοῤῥᾶν κατιέναι]; when between 91° and 180°, that it is descending to the south; between 181° and 270°, that it is ascending towards the south [*sic*, νότον ἀνιέναι]; between 271° and 360°, that it is ascending to the north». Likewise in the tables of the declinations of the sun, under the columns of the degrees of the ecliptic, Theon, I, 144, writes:

From 0° to 90°:	Βοῤῥαῖος καταβαθμός, northern descent.
From 90° to 180°:	Νότιος καταβαθμός, southern descent.
From 180° to 270°:	Νότιος ἀναβαθμός, southern ascent.
From 270° to 360°:	Βοῤῥαῖος ἀναβαθμός, northern ascent.

As in al-Battānī, we read in the *Libro de las taulas alfonsies*, ch. XX (*Libros del saber de astronomia*, vol. IV, p. 136): «et si el logar fuer de un grado fata tres signos. el sol es suuiente en septentrion. et si fuer de tres signos fata .VI. es decendiente en septentrion. et si fuer de .VI. signos fata .IX. es decendiente en mediodía. et si fuer de .IX. signos fata .XII. es suuiente en mediodía» («and if the place is from one degree to three signs, the sun is ascending in the north; and if it is from three signs to six, it is descending in the north; and if it is from six signs to nine, it is descending in the south; and if it is from nine signs to twelve, it is ascending in the south»).[a]

[1] With Ptolemy and al-Battānī the radius is of sixty parts; in our trigonometric formulas, on the other hand, $R = 1$ is taken, so that in our usage the author's rule becomes $\sin\delta = \sin L \sin\varepsilon$.

[a] Nallino's Addenda et emendanda (Part I, p. LXXII) add to this note that Theon's strange definitions are found also in the Astrologia of Vettius Valens, and that Julianus of Laodicea agrees with al-Battānī's usage; the Addenda are translated with Part I's front matter.